

Introduction

CS4201: Information Retrieval and Web Search

Dwaipayan Roy

Department of Computational and Data Sciences

Information Retrieval



Information Retrieval as defined by Gerard Salton in 1960

It is a field concerned with the structure, analysis, organization, **storage**, and **retrieval** of information.

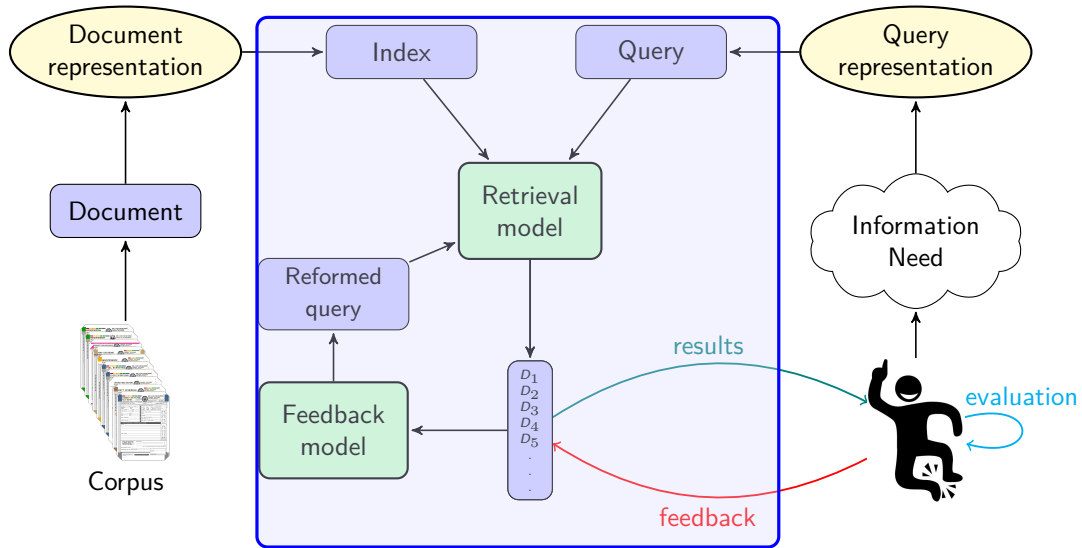
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Information Retrieval

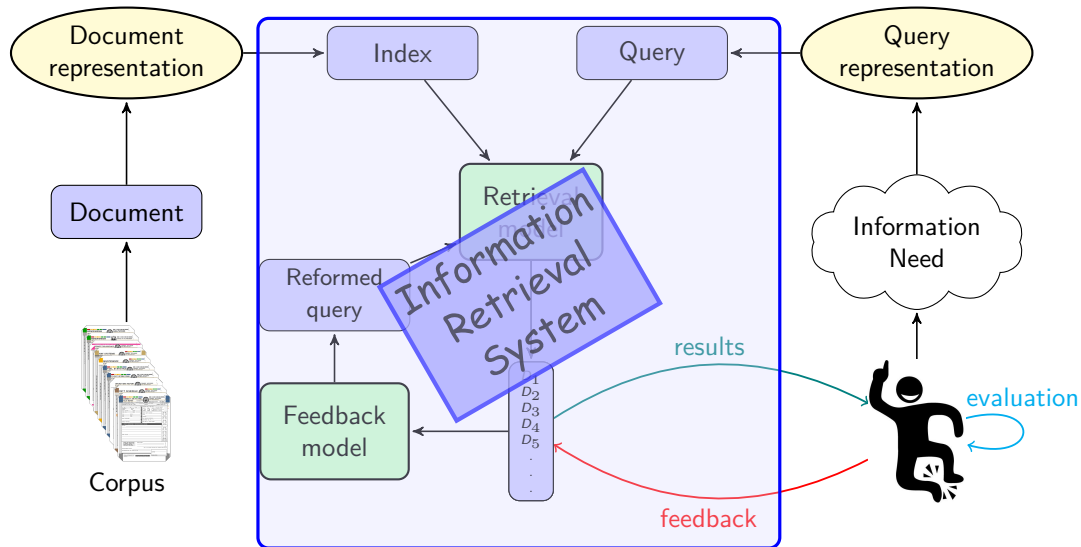
text documents

- **Information Retrieval:** process of seeking out **textual documents** of *unstructured nature*, that satisfies the *information need* (of an user) from within a *collection of documents*.
 - ▶ **Unstructured nature:** no specific structure (*just a collection of words*).
 - ▶ **Information need:** the query of the user (*a set of keywords*).

Information Retrieval: A Graphical Representation



Information Retrieval: A Graphical Representation



The Root: Library Search



Library Search

Typical Queries

Example 1

- Find book(s) with **title** *Macbeth*

Library Search

Typical Queries

Example 1

- Find book(s) with **title** *Macbeth*

Example 2

- Find book(s) **authored** by *William Shakespeare*

Library Search

Typical Queries

Example 1

- Find book(s) with **title** *Macbeth*

Example 2

- Find book(s) **authored** by *William Shakespeare*



Library Search

Complex Queries

Complex Query

- Find the book authored by **Shakespeare** AND with a **character named Desdemona**

Library Search

Complex Queries

Complex Query

- Find the book authored by **Shakespeare** AND with a character named **Desdemona**

Really Complex Query

- Find me the book with the quotation:
 - ▶ ***"To be, or not to be, that is the question"*** OR
 - ▶ ***"There is nothing either good or bad, but thinking makes it so."***

Library Search

Complex Queries

Complex Query

- Find the book authored by **Shakespeare** AND with a **character named Desdemona**

Really Complex Query

- Find me the book with the **quotation**:
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 - “**There is nothing either good or bad, but thinking makes it so.** ”



Library Search

Complex Queries

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Imagine Web Search



Outline

- 1 Information Retrieval: Applications
- 2 Comparing searches in IR and RDBMS

Information Retrieval: A Practical Application



Information Retrieval: A Practical Application



IR is not only Google!

Search Engines: Web Search

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6 DAYS AGO
A water shortage could cut food exports from India, which has emerged as a leading supplier of a number of food products to the world, the water resources minister warned. Water is *kurinattu eera* in the summer months.

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Time to test waters in the boardroom?
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Tamil Nadu water crisis: AIADMK government will go even without
6 DAYS AGO
With Tamil Nadu reeling under a severe drinking water crisis, opposition leader and DMK president M K Stalin on Monday warned that the ruling AIADMK would be outed without an election.

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Q information		
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 Onal2018_Article_NeuralInformationRetrievalAtTh.pdf	1.3 MB	Downloads
 Information Retrieval - A Survey.pdf	1.1 MB	Dropbox/ir/papers
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 Representing Documents and Queries as Sets of Word Embedded Vectors f...	10 items	Dropbox/Apps/ShareLaTeX

Search Engines: Library Search

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word embedding based methods

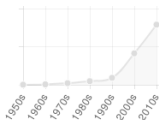
SEARCH

Searched for word embedding based methods [\[new search\]](#) [\[edit/save query\]](#)[\[advanced search\]](#)Searched The ACM Full-Text Collection: 559,764 records [\[Expand your search to The ACM Guide to Computing Literature: 2,854,493 records\]](#) ?

273,887 results found

Export Results: [bibtex](#) | [endnote](#) | [acmref](#) | [csv](#) [6,770 videos found](#)

Result 1 – 20 of 273,887

Result page: [1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#) [9](#) [10](#) [>>](#)Sort by: [relevance](#)**Refine by People**[Names](#) ▶
[Institutions](#) ▶
[Authors](#) ▶
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[Proceeding Series](#) ▶**Refine by Publication Year****1** [Incorporating Prior Knowledge Into Word Embedding for Chinese Word Similarity Measurement](#) [Degen Huang](#), [Jiahuan Pei](#), [Cong Zhang](#), [Kaiyu Huang](#), [Jianjun Ma](#)

April 2018 ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP): Volume 17 Issue 3, May 2018

Publisher: ACM**Bibliometrics:** Citation Count: 0
Downloads (6 Weeks): 7, Downloads (12 Months): 76, Downloads (Overall): 100Full text available: [PDF](#)

Word embedding-based methods have received increasing attention for their flexibility and effectiveness in many natural language-processing (NLP) tasks, including Word Similarity (WS). However, these approaches rely on high-quality corpus and neglect prior knowledge. Lexicon-based methods concentrate on human's intelligence contained in semantic resources, e.g., Tongyici Cilin, HowNet, and Chinese WordNet, ...

Keywords: Chinese word similarity, prior knowledge, word embedding[\[result highlights\]](#)**2** [Query Expansion Using Word Embeddings](#) [Saar Kuzi](#), [Anna Shtok](#), [Oren Kurland](#)

October 2016 CIKM '16: Proceedings of the 25th ACM International on Conference on Information and Knowledge Management

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D. Roy (IISER Kolkata)

1. Introduction

15 / 36

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All Posts People Photos Videos Marketplace Pages Places Groups

Filter results

POSTS FROM

- ☒ Anyone
- ☐ You
- ☐ Your friends
- ☐ Your groups and Pages
- ☐ Public
- [+ Choose a source...](#)

POST TYPE

- ☒ All posts
- ☐ Posts you've seen

POSTED IN GROUP

- ☒ Any group
- ☐ Your groups
- [+ Choose a group...](#)

TAGGED LOCATION

- ☒ Anywhere
- ☐ Kolkata



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Cricketer at ICC - International Cricket Council



Sourav Ganguly
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Sourav Ganguly
PwC India
Works at PwC India
Studied at Vinod Gupta School of Management, IIT Kharagpur '18
 1 mutual friend: Ayan Mukherjee

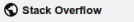
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Q&A for work

Learn More

Search Results

Advanced Search Tips

Ask Question

Results for find text in files in linux

find text in files in linux

Search

500 results

RelevanceNewestMore

4579 votes

42 answers

Q: How do I find all files containing specific text on Linux?

I'm trying to **find** a way to scan my entire **Linux** system for all **files** containing a specific string of **text**. Just to clarify, I'm looking for **text** within the **file**, not **in** the **file** name. When I was ... looking up how to do this, I came across this solution twice: **find** / -type f -exec grep -H '**text-to-find-here**' {} \; However, it doesn't work. It seems to display every single **file in** the system. Is ...

linux text grep directory find

asked Jun 6 '13 by Nathan

724 votes

22 answers

Q: Use grep --exclude/--include syntax to not grep through certain files

I'm looking for the string **foo=** **in text files in** a directory tree. It's on a common **Linux** machine, I have bash shell: **grep -irl "foo=" *** **In** the directories are also many binary **files** which match ... Recurse **in** directories skip **file** matching **PATTERN**. Searching on **grep include**, **grep include exclude**, **grep exclude** and variants did not **find** anything relevant If there's a better way of grepping only **in** ...

unix search shell command-line grep

asked Oct 21 '08 by Piskvor

IR Tasks and Applications

digital library search	desktop search
web search	community Q&A search
enterprise search	federated search
news search	social search
local business search	expert search
image search	product search
video search	patent search
microblog search	recommender systems

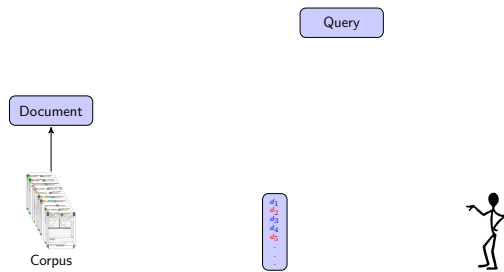
Information Retrieval Experimental Setup

- To compare two retrieval strategies **A** and **B**:
 - ▶ need to use the same collection of documents;
 - ▶ need to search with the same queries.

Information Retrieval Experimental Setup

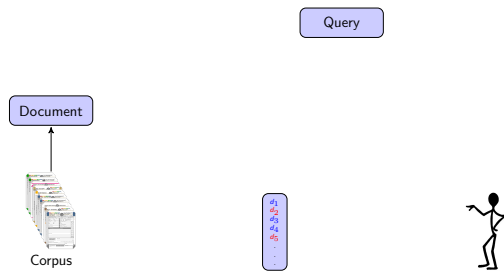
- To compare two retrieval strategies **A** and **B**:
 - ▶ need to use the same collection of documents;
 - ▶ need to search with the same queries.
- Need to have the (overall) same setup.

Ad hoc Information Retrieval: Graphical Representation



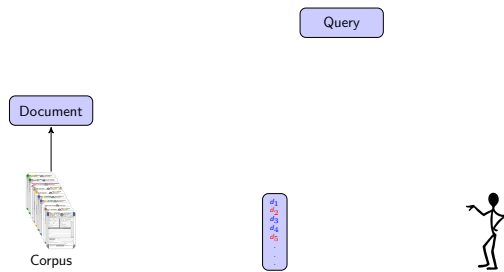
- Given a query and a corpus, find relevant items
 - corpus: a repository of retrievable items;
 - query: a user's expression of information need;

Ad hoc Information Retrieval: Graphical Representation



- Given a **query** and a **corpus**, find **relevant items**
 - ▶ **corpus**: a repository of retrievable items;
 - ▶ **query**: a user's expression of information need;
 - ▶ **relevance**: satisfaction of the user's information need.

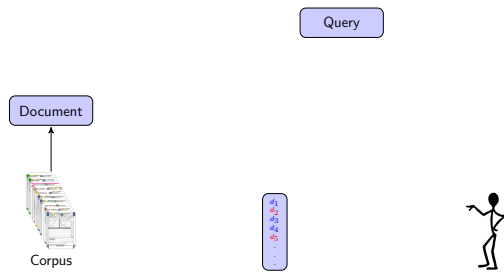
Ad hoc Information Retrieval: Graphical Representation



- Given a query and a corpus, find relevant items
 - corpus: a repository of retrievable items;
 - query: a user's expression of information need;
 - relevance: satisfaction of the user's information need.

- It is called **ad hoc**: the number of queries to test your algorithm is finite here

Ad hoc Information Retrieval: Graphical Representation



- Given a **query** and a **corpus**, find **relevant items**
 - corpus**: a repository of retrievable items;
 - query**: a user's expression of information need;
 - relevance**: satisfaction of the user's information need.

- It is called **ad hoc**: the number of queries to test your algorithm is finite here (infinitely many queries are possible in theory).

Cranfield method (CLEVERDON ET AL., 60s)

Benchmark data

- Document collection: to search into.
- Set of queries / topics: to search with.

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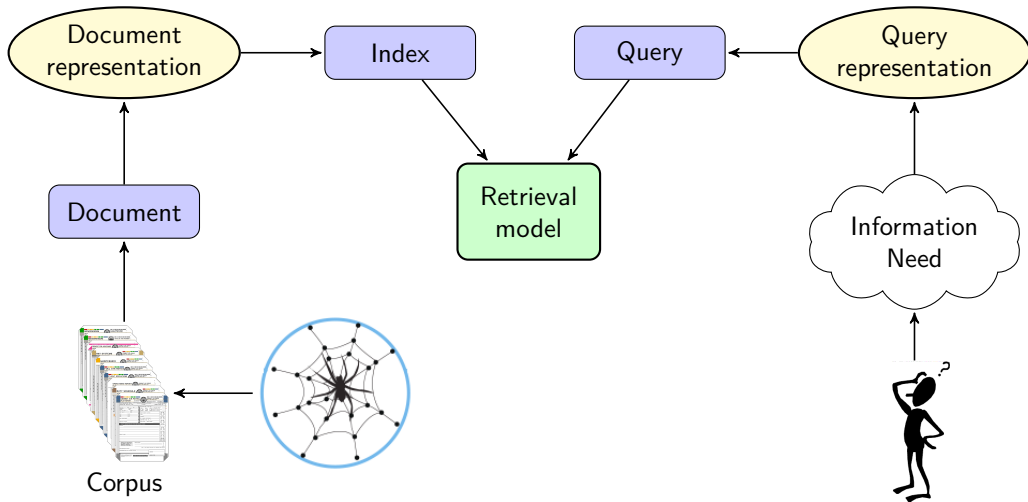
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Web Search: A Graphical Representation



What is crawler?

- The process or program (also called *web spider*, *bot*) used by search engines to automatically download pages from the Web.
- Start with a list of URLs to visit (seed);
- Identify all the hyperlinks in the page and add them to the list of URLs.
- Download all the web pages.

Web crawling

a basic algorithm

Web crawling

a basic algorithm

- initialize a queue Q with the initial set of seed URLs;

Web crawling

a basic algorithm

- initialize a queue Q with the initial set of seed URLs;
- while Q is not empty:

Web crawling

a basic algorithm

- initialize a queue `Q` with the initial set of seed URLs;
- while `Q` is not empty:
 - ▶ `page = Q.remove()`

Web crawling

a basic algorithm

- initialize a queue Q with the initial set of seed URLs;
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 - ▶ `page = Q.remove()`
 - ▶ if page is not HTML/PHP or similar:
 - ★ `continue`

Web crawling

a basic algorithm

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 - ▶ `d = download page`
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Web crawling

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 - ▶ `d = download page`
 - ▶ if error in downloading page:
 - ★ continue
 - ▶ else:
 - ★ parse d
 - ★ put all the URLs in d into Q

Outline

- 1 Information Retrieval: Applications
- 2 Comparing searches in IR and RDBMS

What is Information Retrieval

- **Information Retrieval:** process of seeking out documents of *unstructured nature*, that satisfies the *information need* (of an user) from within a collection of documents.
 - ▶ *Unstructured nature*: No specific structure (just a collection of words).
 - ▶ *Information need*: The query of the user.

Structured (Databases) vs. Unstructured (IR)

- Tends to refer to information in “tables”.

Name	Country	Runs
Sachin	India	18,426
Ponting	Australia	13,704
Jayasuria	Sri Lanka	13,430
Saurav	India	11,363

- Typically allows **range** (for numerical values) and **exact match** (for text) queries;
E.g. `Runs > 12,000 AND Country == India`.

Structured (Databases) vs. Unstructured (IR)

- Typically refers to free-form text.
- Allows:
 - ▶ Term queries including numericals & operators like:
 - ★ find all documents with "221/B Baker Street".
 - ▶ Sophisticated queries like:
 - ★ find all web-pages regarding "betting in Cricket".
 - ▶ ...

Comparison to DBMS

- **Database:** Records
- **Information Retrieval:** Documents

Comparison to DBMS

- **Database:** Records
- **Information Retrieval:** Documents
- Database records (or tuples in relational databases) are typically made up of well-defined fields (or attributes)
- Easy to compare fields with well-defined semantics to queries in order to find matches
- Text is more difficult

Records vs. Documents

- Fetching records:
 - ▶ Find records with `balance > $50,000` in branches of American Express Bank.
 - ▶ Matches easily found by comparison with field values of records.
- Retrieving documents:
 - ▶ Bank frauds in India
 - ▶ Text must be compared to the text of entire news stories.

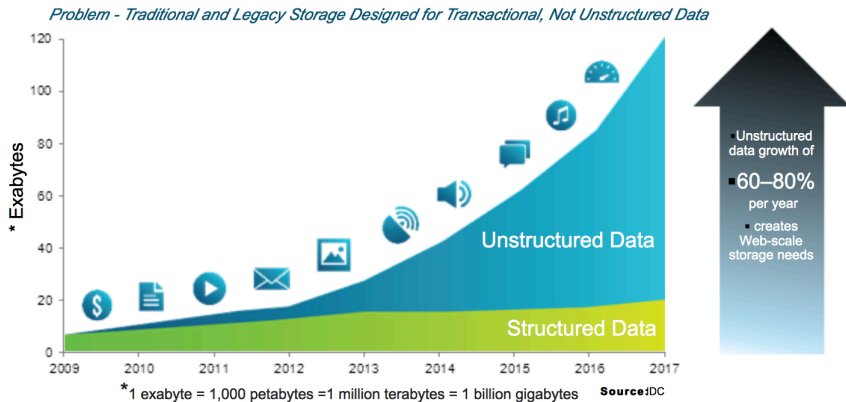
IR vs. DBMS

	IR	DBMS
Query	natural (free-text)	artificial (sql)
Query specification	incomplete	complete
Matching	vague	exact
Items wanted	relevant	all matching
Ranking of items	based on relevance	not applicable

Why are we interested in Unstructured Data!

Why are we interested in Unstructured Data!

Data Growth



Why IR difficult

IR: An uncertain process

- users **don't know** what they **want**
- users **don't know** how to **convey** what they want

Why IR difficult

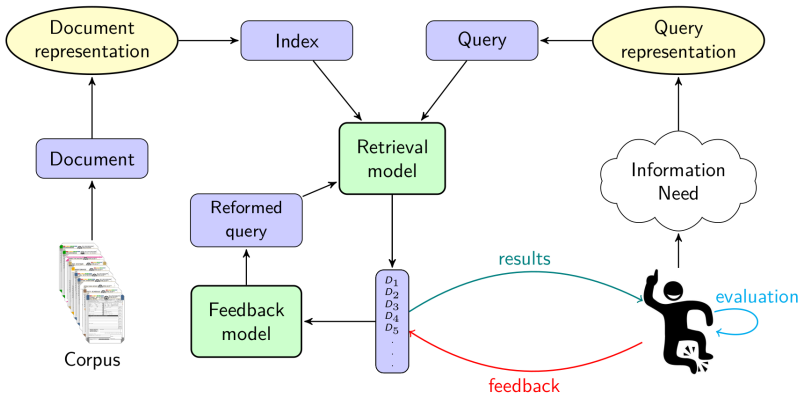
IR: An uncertain process

- users **don't know** what they **want**
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- computers **can't elicit information** like a librarian
- computers **can't understand** **natural language text**

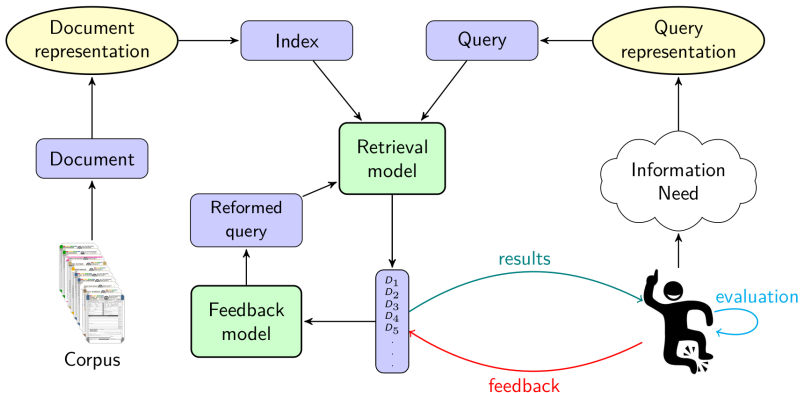
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IR: An uncertain process

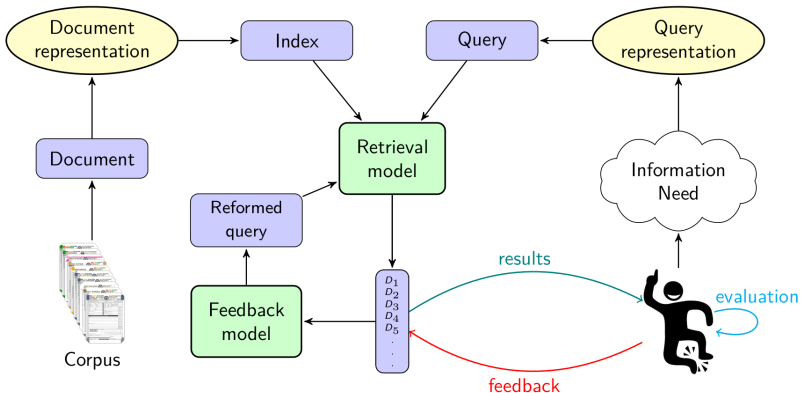
- users **don't know** what they **want**
- users **don't know** how to **convey** what they want
- computers **can't elicit information** like a librarian
- computers **can't understand natural language text**
- the search engine can only **guess** what is **relevant**
- the search engine can only **guess** if a user is **satisfied**



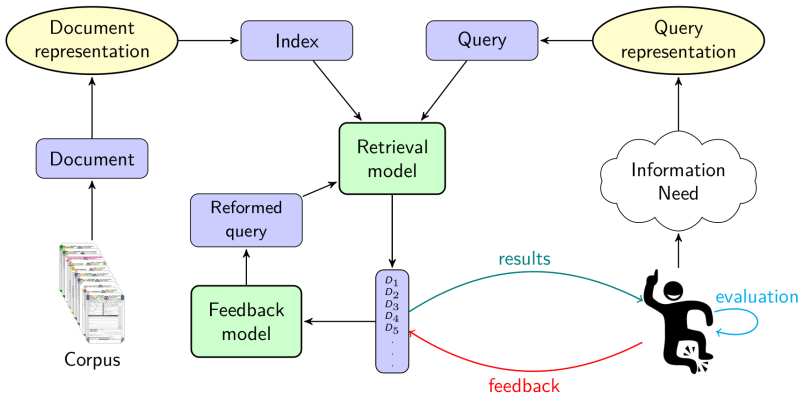
- Document representation
- Indexing
- Query representation
- Retrieval model
- Result evaluation



- Document representation
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- Retrieval model
- Result evaluation
- Feedback model
- Web search



- Document representation
- Indexing
- Query representation
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- Result evaluation
- Feedback model
- Web search
- PyLucene



- Document representation
- Indexing
- Query representation
- Retrieval model (1)
- Result evaluation (1)
- Retrieval model (2)
- Feedback model
- Web search
- PyLucene